charge is $e = 1{\cdot}6 \times 10^{-19}$ coulombs, or $4{\cdot}8 \times 10^{-10}$ electrostatic units, and is therefore exceedingly small. This charge of an atom of electricity is called the *elementary quantum of electricity,* for any electric charge can be only an integral multiple—either positive or negative—thereof.[c] This number and several other important constants of nuclear physics, are given in Table 1 at the end of this book.

We have already pointed out that the ratio of the charge to the mass of electrons, both in electric and magnetic fields, was determined on the basis of the deflection of the cathode rays—of electrons, in other words. Once the magnitude of their charge is known, the electronic mass can be computed. It is only about 1/1,840 of the mass of the hydrogen atom, viz.: $9{\cdot}1 \times 10^{-28}$ gramme. In Table 1 it is designated *rest mass,* because the mass of every body increases with its velocity.

Up till not so very long ago, only electrons with a negative electric charge were known. Positively charged electrons were only discovered during the past decade. Under normal conditions, this positive electron is very short lived; as a rule, it vanishes soon after it comes into being. Otherwise a positive charge also occurs invariably in magnitudes equal to one or more elementary quanta of electricity, in association with masses of the atomic order of magnitude. This fact in itself suggests that the mass of the atom is associated with a positive electric charge which is neutralized by negative electrons, and that ions are produced either by a loss or gain of electrons. But it was still a long way from this concept to the creation of the correct atom model.

## II. RUTHERFORD'S ATOM MODEL

Shortly before the end of the nineteenth century, the way was opened for new decisive developments in nuclear physics. They were introduced by a discovery not directly related to nuclear physics at all: the discovery of x-rays by Wilhelm Röntgen in 1895. The first effect of this discovery was merely the knowledge of a new type of radiation which, although not perceptible directly by human sense organs, could be measured by instruments in the physical laboratory, and which amazed the entire world by its power to penetrate dense layers of matter.

c

In the following year, as a result of his attempt to discover other rays of a similar nature, the Frenchman Henri Becquerel proved that certain substances, notably the compounds of uranium, emitted certain rays of a similar penetrating power, and that this emission took place independently of any external agency. This phenomenon became known as *radioactivity,* and the